



Himanshu Kumar Singh

PROFESSIONAL EXPERIENCE

- **Senior Software Developer** at **BeeHyv Software Solutions** with over **4.5 years** of experience in **AI/ML, Generative AI, Data Science, and Backend Development**.
- Specialized in designing and building scalable **Gen AI applications** using **Python, TensorFlow, PyTorch**, and a range of **open-source AI/ML models**.
- Skilled in developing end-to-end intelligent systems that integrate advanced machine learning techniques with robust backend architectures.

EDUCATION

- Bachelor of Technology from IIT Ropar (2021)

SKILLS SUMMARY

Programming Languages	Python
Frameworks	Fast API, python notebook, Pytorch, TensorFlow, Langchain, Lang graph, Llama-index, Airflow
Database	Postgres, MongoDB, Redshift, Neo4j
DevOps	AWS, GCP, GitHub Action, EKS
Operating System	LINUX, Windows
Version Control	Git, GitHub, GitLab
Others	Backend Development, Gen AI, AI/ML, NLP, Data engineering

PROJECTS

Project Title: GIDR.AI - Data Scientist

Description: Improved chatbot accuracy, enhanced text-to-SQL (txt2sql) performance, and developed document intelligence capabilities.

Responsibilities:

- Designed and implemented evaluation metrics for chatbot content retrieval and response generation.
- Developed txt2sql evaluation metrics and improved accuracy by dynamically selecting few-shot examples from SQL query logs.



- Extracted text from PDFs and images based on user-defined scenarios, highlighting text with bounding boxes.
- Reviewed multiple research papers to benchmark existing methods and improve solutions.
- Built an AI voice agent chatbot capable of updating website DOM elements, allowing users to fill out forms or modify UI components using voice commands. To reduce operational costs, the system can leverage ElevenLabs or Google's built-in Web Speech API.
- Improved document chunk extraction by formatting tables in HTML and aligning images with their corresponding text chunks. The extracted output closely resembles Landing AI's document extraction format. Compared various document extraction methods using LangChain, Docling, Union-Find-based chunking (with business rules), and PyMuPDF.
- Designed and implemented a deterministic state machine workflow, contrasting with the dynamic behaviour of REACT agents.

Technologies Used: GenAI, Python, Fast API, React, GCP cloud Run

Project Title: **AI Agents – EB1A Experts**

Description: Designed and developed an AI agent application to automate operations for the EB1A Experts team and built a public-facing AI client application for end users.

Responsibilities:

- Built an AI agent application to automate content writing for various teams using OpenAI and other LLM providers.
- Developed AI agents to generate and verify content required for EB1A applications.
- Implemented content verification using custom lexical rules with NLTK and named entity recognition with SpaCy and BERT models.
- Automated the CI/CD pipeline and deployed the application on AWS EC2, with integrated logging and monitoring using Grafana, Jaeger, Loki, and related services.
- Developed backend CRUD APIs and implemented AI-driven content generation and verification for a user-facing application.
- Setup of Kong API Gateway with authentication, rate limiting. Handled security related task – trivy scan, sonarqube setup
- Worked on setting up EKS

Technologies Used: Gen AI, Python, Fast API, AWS Services, Grafana stack, Jaeger

Project Title: **Kumbh SahAIyak**



Description: Designed and developed Kumbh SahAIyak RAG chat App with guardrail and deep face model of audio and video application.

- Built a robust question-answering pipeline by combining BM25 with fuzzy search and vector search for initial retrieval and applying BGE Reranker to optimize relevance and ranking accuracy.
- Prompt finetuning with textgrad for RAG application and performed bge embedding fine-tuning to enhance model retrieval and reasoning capabilities.
- Designed and enforced AI governance by mimicking AWS Guardrails, building custom models and rule-based systems to ensure content safety, compliance, and reliability.
- Developed an AI-driven deepfake system for CM video generation, integrating advanced audio cloning technologies including Eleven Labs, Coqui-TTS, VITs and NaturalSpeech2 for high-fidelity voice synthesis. Implemented precise lip-syncing using Wav2Lip, achieving realistic and synchronized video outputs.
- Fine-tuning VITs and Coqui-tts model. Cleaning the audio data using Nemo NVidia and using threshold and filter-based technique. Also testing it with Speech to text using Bhashni and OpenAI whisper model.

Technology involved: Python, Python notebook, PyTorch, docker.

Project Title: [EasyKnock](#)

Description: EasyKnock is a real estate technology company that provides homeowners with innovative solutions to access home equity without taking on new debt or relocating. Their services, such as Sell & Stay and Moveability, offer financial flexibility by allowing homeowners to sell and lease back their homes or access funds to buy a new home before selling their current one.

Responsibilities:

- Build data pipeline that fetches data from third party business like housecanary, Clear Capital, ClearEdge, from sales teams who uses salesforce and AirTable, from online advertising platform like google AdWords, Facebook, for Machine learning workflow and from different internal micro-services to sync data for Analyst and Business Entity.



- Migration and Deployment of Machine learning and airflow related code base from GCP to AWS.
- Deployment and testing of Airflow Code and DBT code using GitHub Action
- Addressed reported bugs from customers or QA and resolved issues pertaining to database entries.

Technologies involved: Python, Airflow, GitHub Actions, Docker, Kubernetes, AWS (VPS, EKS, CloudWatch, lambda, API Gateway, S3, Redshift, Document DB, Postgres, SageMaker, MWAA), GCP (Cloud-Function, Bigquery, Cloud Storage, Cloud Composer)

Project Title: AI-Evaluator

Description: AI-Evaluator is an in-house trainee assistance system that quizzes trainees on topics they have completed, evaluates their responses, and assigns scores using large language models (LLMs).

Responsibilities:

- Created Backend APIs using Fast API that supports Authentication, generating and grading question using LLMs
- Integrated in Postgres and deployed the code as docker-compose file that has Postgres, backend and Frontend as a service.

Technologies involved: Python, Fast API, Docker Compose, LLMs

Project Title: LTIMindtree - FIU-IND

Description: The Financial Intelligence Unit of India (FIU-IND) is the central national agency responsible for receiving, processing, analysing, and disseminating information relating to suspect financial transactions to combat money laundering and terrorist financing. It operates under the Ministry of Finance and plays a crucial role in ensuring the integrity of the financial system in India.

Responsibilities:

- Built the application Data Quality Report in python framework. It processes batch data from 17 different types of financial institution like Banks, exchange house etc in different format and checks the quality of data based on 32 rules like duplicate reports, unique id validation rules, external validation of attributes from 8 different institution via API, ML rules etc.
- Hands-on Data Modelling and View creation involving multiple tables for consumption of user interface, network visualization using Neo4j tool.
- Implementation of Risk score, Data preprocessing and EDA using python library, SQL queries.



- Continuously updated and maintained the automation suite to align with evolving data source technologies.

Technologies involved: Python, Pandas, SkLearn, Neo4j, MS SQL Server

Project Title: **Armour Design with Deep learning**

Description: Armour used by Army to protect against guns are often heavy and are made up of many layers of composite. Our aim was to find optimum layer of composite and reduce the time of simulation using deep learning. Data was generated using LS-DYNA simulation

Responsibilities:

- Our team applied LSTM and neural network model on simulation datasets of Armor bullet impact. Through the literature survey, we found researchers have used only simple neural networks.
- Our findings focus on creating a lesser number of simulations but utilizing the time-series intermediate simulation data on LSTM, unlike simple neural networks.
- It is timesaving and computationally inexpensive.

Technologies involved: Python, Pandas, SkLearn, TensorFlow

Project Title: **E-commerce Backend**

Description: Designed and developed an e-commerce backend using Go and the Gin framework, supporting core shopping features.

Responsibilities:

- Implemented user authentication and login functionality with secure session handling.
- Developed shopping cart features, including adding, updating, and removing products.
- Designed RESTful APIs to enable integration with frontend applications.

Technologies: Go, Gin, MongoDB, REST APIs

Project Title: **Satyagraha – Tech Contributor (Volunteer)**

Description: Contributed to the development of core web pages and deployment pipelines for Satyagraha's digital platform.

Responsibilities:



- Designed and developed the landing page and authentication page for the Satyagraha website.
- Implemented responsive UI using React, CSS, and HTML.
- Configured cloud deployment workflows using Google Cloud Platform (GCP) and automated CI/CD pipelines with GitHub Actions.

Technologies: React, CSS, HTML, GCP, GitHub Actions